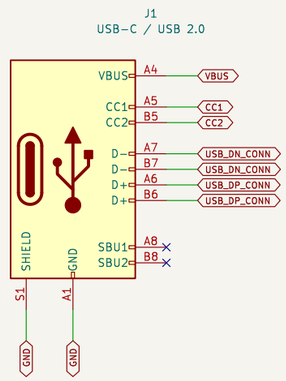
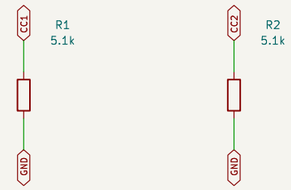
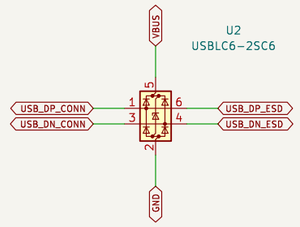
# POWER & USB | 5 V USB-C sink, native USB programming



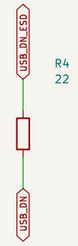



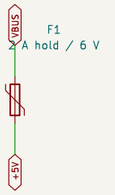






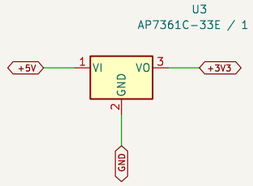




22 ohm resistors near MCU.
Route D+/D− as a 90 ohm differential pair.
ESD at connector; shell bonded to GND.



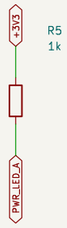










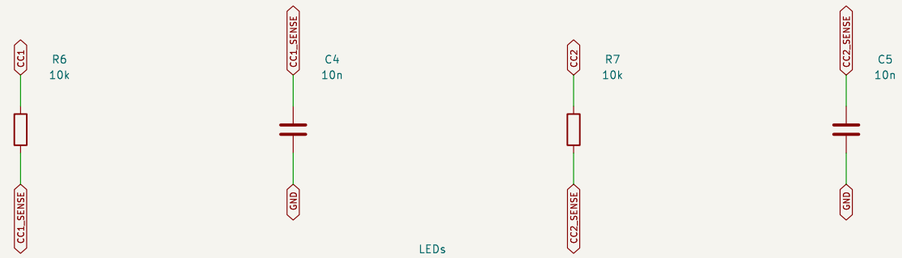









POWER POLICY
Use a regulated 5 V / 3 A USB-C source for full output. No USB PD.
Firmware must inspect CC1/CC2 and USB configuration before raising load.
Default current mode: LEDs and amplifier off; restrict radio activity.
LDO dissipation: 0.51 W at 300 mA; verify temperature in the enclosure.
Large LED capacitor is isolated by U6 until firmware enables soft start.




MCU

File: mcu.kicad_sch

LEDs

File: leds.kicad_sch

Audio

File: audio.kicad_sch

Inputs & Expansion

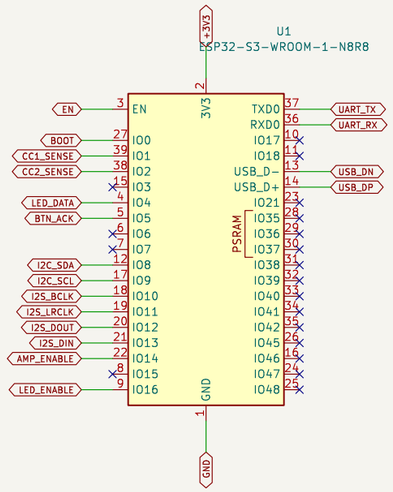
File: inputs.kicad_sch

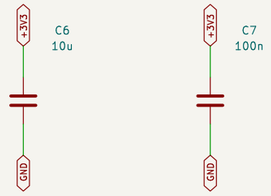
Prototype schematic; PCB placement study is not fabrication-ready
Power & USB
**HackNC 2026**

Sheet: /
File: clearday-home.kicad_sch

***Title: Clearday Home***

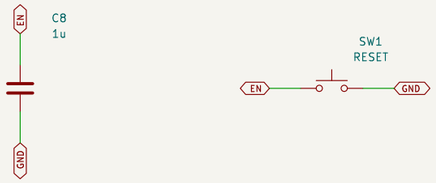
| Size: A3 | Date: 2026-10-11 | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 8.0.9 | | Id: 1/5 |










GPIO ASSIGNMENTS
GPIO1 / GPIO2 CC1_SENSE / CC2_SENSE (ADC1)
GPIO4 LED_DATA -> U5 level shifter
GPIO5 BTN_ACK, active low
GPIO8 / GPIO9 I2C_SDA / I2C_SCL
GPIO10 / GPIO11 I2S_BCLK / I2S_LRCLK, MCU master
GPIO12 / GPIO13 I2S_DOUT to amp / I2S_DIN from mic
GPIO14 AMP_ENABLE (high selects stereo mix)
GPIO16 LED_ENABLE (slow-start switch)
GPIO19 / GPIO20 USB_DN / USB_DP, native USB
GPIO43 / GPIO44 UART_TX / UART_RX, 3.3 V logic

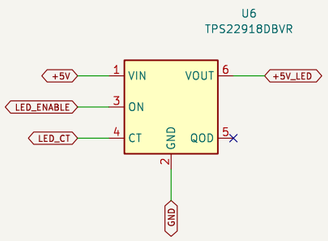
Hold BOOT, tap RESET, release BOOT to enter ROM loader.
EN: 10k / 1u RC, approximately 10 ms.
All module GND pads, including exposed pad, connect to GND.

RESERVED PINS
GPIO0: BOOT only; external pull-up.
GPIO3, GPIO45, GPIO46: strapping pins, no loads.
GPIO35, GPIO36, GPIO37: octal PSRAM, no connection.
Do not place copper or components in antenna keepout.
Use a 15 mm clearance beyond the antenna where possible.
Keep LED data and AMP_ENABLE low during startup.

Prototype schematic; PCB placement study is not fabrication-ready
MCU
**HackNC 2026**

| | | |
|---|---|---|
| Sheet: /MCU/ | | |
| File: mcu.kicad_sch | | |
| ***Title: Clearday Home*** | | |
| Size: A3 | Date: 2026-10-11 | **Rev: A** |
| KiCad E.D.A. 8.0.9 | | Id: 2/5 |

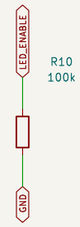
# LEDs | 12-pixel RGB ring, switched 5 V

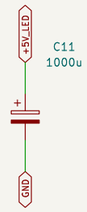




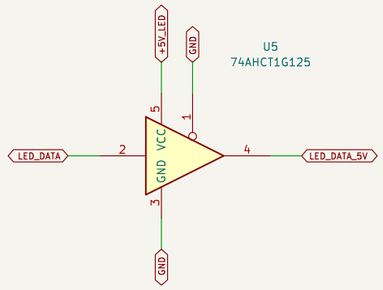


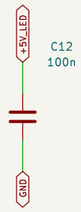


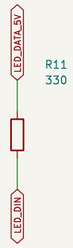



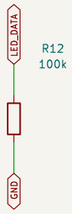



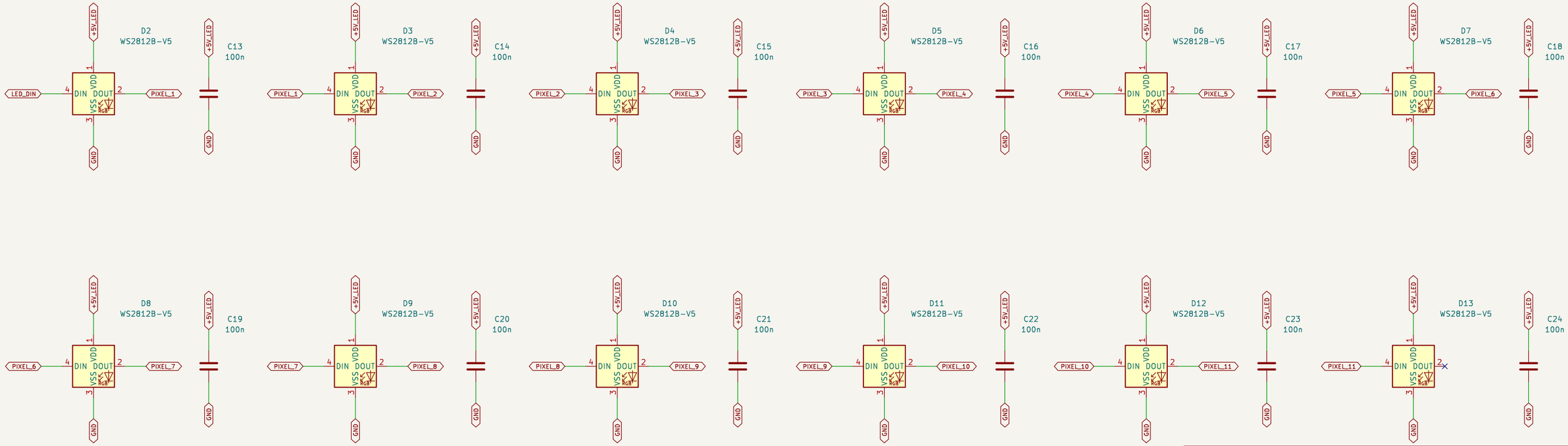

CT = 10 nF: about 26 ms rise at 5 V.
1000 uF charging current: about 0.2 A.
QOD open; no forced capacitor discharge.
Allow >= 50 ms before transmitting LED data.
Keep LED_DATA low when U6 is off.



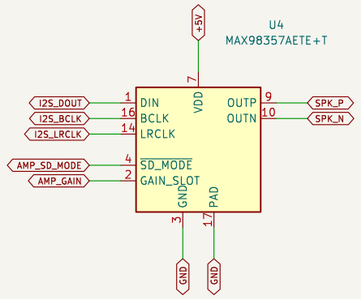

D2 -> D13 clockwise on PCB. Last DOUT is intentionally unused. One 100 nF capacitor per pixel.

Prototype schematic; PCB placement study is not fabrication-ready
LEDs
**HackNC 2026**
Sheet: /LEDs/
File: leds.kicad_sch

***Title: Clearday Home***

| | | |
|---|---|---|
| Size: A3 | Date: 2026-10-11 | **Rev: A** |
| KiCad E.D.A. 8.0.9 | | Id: 3/5 |

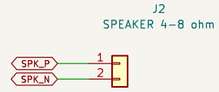
AUDIO | Full-duplex I2S; mono speaker, left-channel microphone

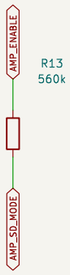


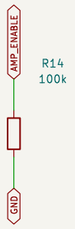


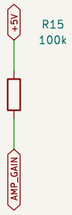




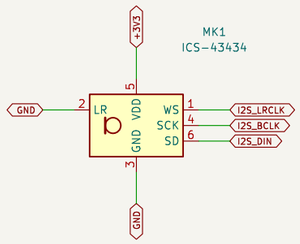



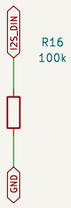




SD_MODE: 560k from 3.3 V GPIO into internal 100k.
Nominal 0.500 V selects (L + R) / 2.
GPIO low mutes. External 100k holds enable low at reset.
GAIN_SLOT: 100k to VDD selects +3 dB.
Never connect either speaker terminal to ground.
No headphone or grounded line-output connection.










MCU supplies BCLK and LRCLK to both devices.
48 kHz stereo frames, 32 bits per slot: 3.072 MHz BCLK.
Microphone outputs 24-bit data in left slot (LR = GND).
100k SD pulldown defines the unused high-Z slot.
No MCLK needed. Decouple directly at each IC.
Do not obstruct the microphone acoustic port.

Prototype schematic; PCB placement study is not fabrication-ready
Audio
**HackNC 2026**

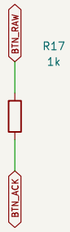
Sheet: /Audio/
File: audio.kicad_sch

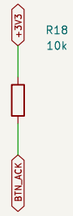
***Title: Clearday Home***

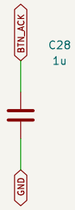
| Size: A3 | Date: 2026-10-11 | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 8.0.9 | | Id: 4/5 |

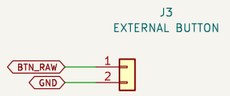
# INPUTS & EXPANSION | Acknowledge button, I2C and UART



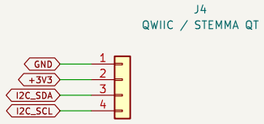



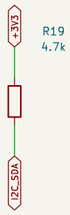



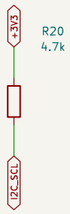




Active low. 1k limits capacitor discharge current.
Press time constant: 0.9 ms; low settles near 0.3 V.
Release time constant: 10 ms. Firmware debounce: 30 ms.
External button shorts pins 1 and 2 only.

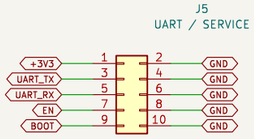









Qwiic: 1 GND, 2 3V3, 3 SDA, 4 SCL. 3.3 V only. Sensor budget <= 50 mA.
Start at 100 kHz. Remove onboard pull-ups if parallel sensor pull-ups are too strong.
Service header: TX/RX names are from the MCU perspective. Use a 3.3 V UART adapter.
EN and BOOT require open-drain drive. J5 is not a standard ARM debug connector.

Prototype schematic: PCB placement study is not fabrication-ready
Inputs & Expansion
**HackNC 2026**

| | | |
|---|---|---|
| Sheet: /Inputs & Expansion/ | | |
| File: inputs.kicad_sch | | |
| ***Title: Clearday Home*** | | |
| Size: A3 | Date: 2026-10-11 | **Rev: A** |
| KiCad E.D.A. 8.0.9 | | Id: 5/5 |